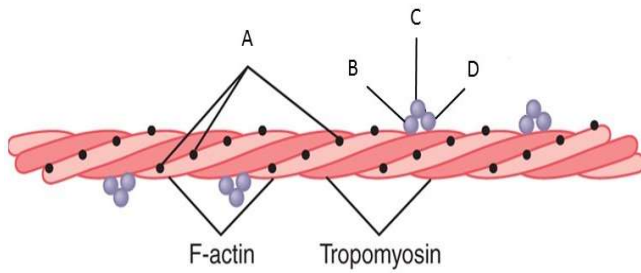


1-



In the opposite figure, which of the letter points to

"troponin C"?

- ✓ C
- B
- D
- A

2- Which of the following is of great importance in onset of contraction in a smooth muscle fiber?

- Ca binding to troponin
- Active sites exposure
- ✓ Myosin light chain kinase activation
- Frequent action potentials

3- Which of the following best describes the cause of slow cycling of the myosin cross-bridges in the smooth muscle fibers?

- Tropomyosin coverage on actin filaments
- ✓ Far less ATPase activity
- Less cAMP cycling
- Inositol triphosphate involvement makes it long.

4- Regarding the agent of transmission in different kinds of synapse, which of the following is correct?

- Ions transmit the signals in a chemical synaps.
- As a matter of fact, presynaptic vesicles act as important agents in both synapses.
- ✓ In this case, ion current plays a major role in an electrical synapse.
- The role of voltage gated Na channeles should not be ignored in a chemical synaps.

5- Which of the following muscle is controlled just by nerve signals?

Small intestine muscle

Bile duct muscle

Arteries

✓ Iris muscle of the eye

6- How does an action potential penetrate into the depth of a skeletal muscle fiber?

Due to the existence of dihydropyridine receptors in the membrane as a main factor

The crucial role of extensive sarcoplasmic reticulum should not be ignored.

Ryanodine receptors help action potential reach there.

✓ Trough the transverse tubules

7- The most important advantage of an electrical synapse is

✓ rapid and synchronous firing of interconnected cells

amplifying signals

shortening the duration of action potential

the presence of gap junctions

8- Which of the following refers to similarity between skeletal and smooth muscle fibers?

The same molecular structure of actin filaments in both

✓ Need for ATP for contraction in both

Striated appearance in both

Existence of dense bodies in both

9- Rigor mortis occurs when contraction cycle stops in

Step 2

✓ Step 3

Step 4

Step 1

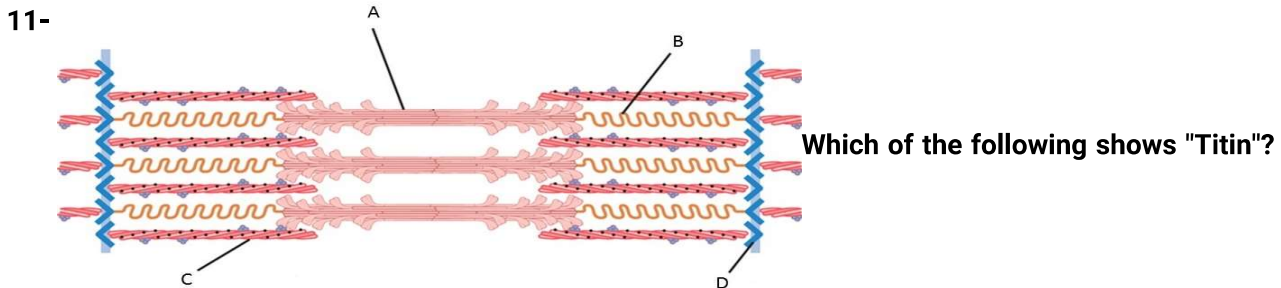
10- Which of the following exactly causes the vesicles to fuse with the presynaptic membrane?

Reaching the action potential at the synapse

✓ The rise in intracellular Ca concentration

Opening of active zones

Opening of voltage gated Na channels



D

✓ B

C

A

12- The most Ca ions required for skeletal muscle contraction is supplied by

Extracellular fluid

✓ Sarcoplasmic reticulum

Internal environment

Sarcoplasm

13- Which of the following has to occur for smooth muscle relaxation?

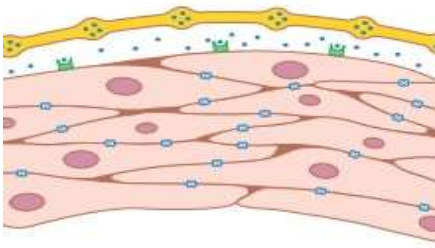
When vesicle release comes to a stop

Ryanodine receptors activation

Bringing action potential to a stop

✓ Removal of Ca ions from the intracellular fluids

14-



What does this figure show?



Neuromuscular junctions in unitary smooth muscle

Neuromuscular junctions in skeletal muscle

Neuromuscular junctions in multi-unit smooth muscle

Release of transmitters no matter how to innervate

15- Which of the following refers to a receptor that gates ion channels directly?

Serotonin receptors at synapses in the cerebral cortex



Nicotinic acetyl choline receptor at the neuromuscular junction

Norepinephrine receptors at synapses in the cerebral cortex

Metabotropic receptors in the whole body

16- Excitation-contraction coupling in a skeletal muscle fiber means

how an action potential in a motor neuron results in an action potential in a muscle fiber

how an action potential causes opening of voltage gated Na channels



how an action potential travelling along the membrane leads to actin- myosin interaction

acetyl choline release in the neuro muscular junction

17- Which factor causes detachment of the myosin head from the actin filaments in the skeletal muscle fibers?

ATPase activity of the myosin head

Release of ADP and phosphate that were previously attached to the head

Tilting the heads toward the arms



Binding of new ATP

Embryology

18- The amniotic cavity forms during the process of implantation of the blastocyst. The amniotic cavity forms within which of the following structures?

- ✓ Epiblast
- Cytotrophoblast
- Syncytiotrophoblast
- Maternal endometrium

19- Which of the following develops from the multinucleated syncytium, the syncytiotrophoblast and serves as the beginnings of gas exchange between mother and embryo?

- Bilaminar disk
- Amniotic cavity
- Cytotrophoblast
- ✓ Lacunae

20- Maternal serum alpha-fetoprotein increases if the fetus has

- oligohydramnios
- Ladd's bands
- ✓ anencephaly
- Bochdalek hernia

21- The outer layer of the umbilical cord is continuous with, and similar in structure to the

- chorionic membrane
- ✓ amniotic membrane
- cytotrophoblast
- syncytiotrophoblast

22- Which of the following cells form endoderm and mesoderm?

- Non-migrating hypoblast cells
- Non-migrating epiblast cells
- Migrating epiblast cells
- Migrating hypoblast cells

23- Syncytiotrophoblast

- ✓ secretes chorionic gonadotropin
- gives rise to the decidua basalis
- begins to develop on the 4th day postfertilization
- degenerates once cytotrophoblast takes over its function

24- _____ Changes occur in it that inhibit entry of sperm

- Acrosome
- Pronuclei
- Capacitation
- ✓ Zona pellucida

25- Implantation of the blastocyst cannot occur without prior

- formation of the chorion
- fragmentation of the primary yolk sac
- ✓ hatching from the zona pellucida
- differentiation of syncytiotrophoblast

26- Cells in somites give rise to

- the mandible
- ✓ ribs
- sympathetic ganglia
- dorsal root ganglia

27- The zona pellucida plays an important role in

- neural crest migration
- embryonic blood production
- ✓ preventing premature implantation
- the formation of the prochordal plate

28- Somites

- are the source of cells in the nucleus pulposus
- ✓ are derived from the paraxial mesoderm
- give rise to the cartilages of the branchial arches
- give rise to cardiac muscle

29- Which of the following structures are derived from the neural crest

- sclerotomes
- ✓ sympathetic ganglion
- cells of the adrenal cortex
- somatic motor neurons

30- The nucleus pulposus is the embryological remnant of which of the following?

- Neural plate
- Sclerotomes
- ✓ Notochord
- Neural tube

31- Which of the following is associated with oligohydramnios

- tracheo-esophageal fistula
- anencephaly
- ✓ renal agenesis
- myelomeningocele

32- Phenomena that creates female gametes, oogenesis is ceased at prophase I up till puberty. This describes the DNA content of a female's gametes during her childhood.

- 23 chromosomes, 46 chromatids
- 23 chromosomes, 23 chromatids
- ✓ 46 chromosomes, 92 chromatids
- 46 chromosomes, 46 chromatids

33- Amniotic fluid volume can be expected to rise above normal if the fetus

- ☐ has renal agenesis
- ☒ has esophageal atresia
- ☐ has Hirschsprung's disease
- ☐ has a Bochdalek hernia

34- The primitive streak

- ☐ gives rise to the allantois
- ☐ first appears in the roof of the amniotic cavity
- ☐ gives rise to ectomesenchyme of the 1st branchial arch
- ☒ is derived from epiblast

35- Which two embryological structures together form the bilaminar disk?

- ☐ Cytotrophoblast and Hypoblast
- ☐ Epiblast and Cytotrophoblast
- ☒ Hypoblast and Epiblast
- ☐ Epiblast and Embryoblast

36- During this phase, the nuclear envelope begins to disappear

- ☐ S phase
- ☒ Prometaphase
- ☐ cytokinesis
- ☐ Anaphase

37- _____ Haploid nuclei that fuse to form a zygote

- ☐ Capacitation
- ☐ Acrosome
- ☒ Pronuclei
- ☐ Polar body

38- All of the following structures are necessary for blastocyst implantation EXCEPT

zona pellucida

✓ cytotrophoblas

syncytiotrophoblast

endometrium in progestational stage

39- When does formation of primary spermatocytes begin?

month 5 of infancy

✓ puberty

week 4 of embryonic life

birth

40- Chemotherapy treatment often targets mitotically active cells. Although this treatment is beneficial for cancer, it can negatively effect implantation and embryonic growth. Which of the following layers would be most affected by a slowed cell division rate?

Syncytiotrophoblast

✓ Cytotrophoblast

Hypoblast

Amnioblast

41- Which week of embryonic development *usually* corresponds to the first missed menstrual period?

✓ Week three

Week two

Week four

Week one

42- When does a primary oocyte complete its first meiotic division?

at 5 months of fetal life

at fertilization

after the ovulatory surge of FSH and LH

✓ at ovulation

43- Which of the following cells form endoderm and mesoderm?

- ☐ Migrating hypoblast cells
- ☒ Migrating epiblast cells
- ☐ Non-migrating epiblast cells
- ☐ Non-migrating hypoblast cells

44- In a growing embryo, the vertebrae encase notochord and neural tube in a solid bone like structure. Vertebrae are formed from which of the following embryological structures?

- ☐ Myotomes
- ☒ Primaxial domain
- ☐ Sclerotomes
- ☐ Dermatomes

Histology

45- Which of the following Histological Structure Is Observed Regular Dens Connective Tissue?

- ☒ Perichondrium Tissue
- ☐ Matrix
- ☐ Perilacuna
- ☐ Articular Cartilage

46- Russel bodies are characteristic for which cell?

- ☒ plasmacell
- ☐ kupfer
- ☐ Adipocyte
- ☐ mastcell

47- which one is dense regular connective tissue?

- ☐ Dermis of skin
- ☒ Tendon
- ☐ Adipose tissue
- ☐ Areolar tissue

48- in which cell we have major basic protein?

- ✓ Eosinophil
- monocyte
- lymphocyte
- macrophage

49- In which Type of the Cartilage Tissue is observed Collagen Type I?

- Elastic
- ✓ Fibro
- Hyalin
- Nothing

50- The type of epithelial tissue consist of many nuclei are not in the same level called:

- stratified non-keratinized squamous
- ✓ pseudostratified
- stratified columnar
- stratified cuboidal

51- Which one of these choices is more important to distinguish of Perikaryon?

- cell processes
- amount of cytoplasm
- ✓ The nucleus
- Nissl bodies

52- In which Types of Cells is observed In Metaphysis Plate?

- Osteoclast
- ✓ Chonderoblast
- Fibroblast
- Chonderoclast

53- Which layer of the epidermis is thickest and has polyhydral cells?

- ✓ Stratum spinosum
- Stratum basale
- Stratum corneum
- Stratum granulosum

54- In which part of the Mature bone is observed Osteoclast?

- ✓ Endosteum
- Periosteum
- Epiphysis Plate
- End Plate Bone

55- Red blood cell is because it has In the wall.

- fragile-spectrin
- flexible-fibrinogen
- ✓ flexible-spectrin
- fragile-actin

56- In which muscle types nucleus have flat shape?

- striated muscle and cardiac muscle
- cardiac muscle and skeletal muscle
- ✓ smooth muscle and striated muscle
- cardiac muscle and smooth muscle

57- Which type of junctions is for adhesion of cell to cell?

- ✓ Desmosomes
- Tight J
- Gap J
- Hemidesmosomes

58- Which of the following cells is involved in Blood- Brain-Barrier?

Ependymal cells-Microglia

Oligodendrocytes- Sattelite

Schwann cells- Perimyisum



Astrocytes- endothelial cells

59- Which of the glial cell formed perivascular feet?

Schwann cells



Astrocytes

Oligodendrocytes

Microglia

60- What type of glands are the ceruminous glands in ear?

Endocrine gland



Apocrine gland

Sebaceous glands

Eccrine gland

61- Which type of regulatory protein located in M or Z line of the skeletal muscle?

Neboulin- Actin

Actinin - Myosin

Titin- Desmin



Myomesin- α actinin

62- Identify the INCORRECT statement:

Microvilli increase the surface area of the epithelia cells.

Pseudostratified epithelium lines respiratory system.



Covering epithelium are classified by the type of cells.

The basement membrane consists of: the reticular and the basal lamina.

63- In which Type of the Bone Tissue is observed Regular Haversian System ?

Spongy Bone

✓ Compact Bone

Nothing

Alveolar Bone

64- What are the basophilic clusters of ribosomes and RER in neurons called?

Perikaryon

Ganglion

Terminal button

✓ Nissl bodies

65- following diapedesis only type of leukocytes can return from the tissues back to the blood. Which one is this?

basophil

✓ Lymphocyte

neutrophil

monocyte

66- In which Types of Cells is observed In Iregular Haversian System?

Osteoblast

Chonderoclast

✓ Osteocyte

Osteoclast

67- What is the name of antigen presenting cell in skin?

Keratinocyte

Merkel

Melanocyte

✓ Langerhans

68- in which cell we have major basic protein?

- ✓ Eosinophil
- monocyte
- lymphocyte
- macrophage

69- structure of framework of spleen and lymphnodes are composed of which fibers?

- collagen bundles
- elaunin
- elastic
- ✓ reticular

70- which cell has fewest number in the normal blood smear?

- Eosinophil
- Monocyte
- ✓ Basophil
- Neutrophil

71- What type of cells are the most part of skin epidermis and hypodermis?

- Melanocyte- Fibroblast
- ✓ Keratinocyte - Adipocyte
- Macrophage- Merkel
- Fibroblast- Keratinocyte

72- Which of the following is the most characteristic type of intermediate filament protein found in muscular cell?

- Cytokeratin
- Integrin
- ✓ Desmin
- Cadherin

73- These cells surround myelinated and non myelinated axons that provide the electrical insulation

- ✓ Schwann cell
- Oligodendrocytes
- Fibrous astrocytes
- Protoplasmic astrocytes

74- In which part of the Cartilage Tissue is observed Chondroblast Like?

- Periosteum
- ✓ Perichondrium
- Lacuna
- End Plate

75- What do you call the simple squamous epithelium that lines the wall of vessels?

- Pseudostratified
- Mesothelium
- Epithelioid
- ✓ Endothelium

76- heat production is the most important function in which tissue?

- tendon
- ✓ brown adipose tissue
- areolar tissue
- dermis

77- Hydroxyapatite crystals near which cell has a natural deposition and by which ion is involved in cellular responses?

- Mesenchymal – H^+
- ✓ Osteoblast - H^+
- Osteocyte – Ca^{2+}
- Fibroblast – Ca^{2+}

78- In which Type of the Joint Tissue is observed Synovial Membrane?

- ✓ Diarthrosis Joint
- Synchondrosis Joint
- Synostosis Joint
- Syndesmoses Joint

79- What is the following is INCORRECT?

- cardiac muscle and smooth muscle nucleus located in center part
- cardiac muscle and skeletal muscle have sarcomere
- Striation was seen in longitudinal section of cardiac muscle and skeletal muscle
- ✓ Striation was seen in transverse section of cardiac muscle and skeletal muscle

80- This type of neuron consist of many dendrites and one true axon, and are more:

- ✓ Multipolar
- Bipolar
- Unipolar or pseudo unipolar
- Apolar